

SOHAIB AKHLAQ

AI Engineering — Full-Stack Development — Software Engineering

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🐙 GitHub in LinkedIn </> LeetCode 🧑 Portfolio

Professional Summary

Software Engineering student at FAST NUCES (CGPA: 3.5/4.0) specializing in AI engineering, LLM-powered automation systems, and full-stack intelligent platforms. Built production-ready AI workflow automation systems using FastAPI, Sentence Transformers, Groq LLM APIs, vector-based retrieval, and modern frontend architectures. Actively learning scalable AI systems with REST APIs, analytics dashboards, automated testing, and deployment pipelines. Developing foundational skills in machine learning workflows, NLP pipelines, and software engineering best practices.

Education

National University of Computer & Emerging Sciences (FAST NUCES)

Islamabad, Pakistan

Bachelor of Science in Software Engineering

Expected May 2028

CGPA: 3.5/4.0

4th Semester

Relevant Coursework: Data Structures & Algorithms, OOP, Database Systems, Operating Systems, Artificial Intelligence, Software Design & Architecture, Software Requirements Engineering, Fundamentals of Software Engineering

Technical Skills

Languages: Python, JavaScript, TypeScript, C++, Java, SQL, HTML5, CSS3

AI / ML & LLM Engineering: LLM Integration (Groq, Gemini, OpenAI APIs), LangChain, Agentic AI Systems, FastAPI, Sentence Transformers, NLP Pipelines, Prompt Engineering, Vector Search & Embeddings, Zero-shot Classification, Hugging Face, Sentiment Analysis, HL7 FHIR, AI Workflow Automation

Full-Stack Development: Angular 17, React.js, Node.js, Express.js, MongoDB, REST APIs, JWT, RBAC, Socket.io, Tailwind CSS, Chart.js

DevOps & Tools: Docker, Nginx, GitHub Actions CI/CD, Git, Linux, Vercel, Render, VS Code

Testing & QA: Postman, Selenium, Jest, Katalon Recorder, Manual Testing, Thunder Client, Jira

Experience

Lab Demonstrator – Object-Oriented Programming (Paid)

Jan 2026 – June 2026

FAST-NUCES, Islamabad, Pakistan

- Assisted 50+ students per session with Java debugging, OOP concepts (inheritance, polymorphism, design patterns), and lab exercises; improved lab completion rates by **25%** through targeted one-on-one mentoring.
- Collaborated with faculty on lab design, evaluations, and technical curriculum support across multiple course iterations.

Projects

DischargeAI – Autonomous Clinical Ward Handover Agent

Python, Groq LLM (llama-3.3-70b), HL7 FHIR,

FastAPI, JavaScript

github

- Engineered an autonomous AI agent that transforms unstructured clinical notes into structured discharge summaries, GP letters, and HL7 FHIR bundles – eliminating manual documentation overhead for ward clerks and clinicians.
- Designed a rule-based escalation framework handling **80% of cases autonomously** while detecting edge cases (medication conflicts, high-risk social flags, critical lab results) and routing them for mandatory human review.
- Implemented zero-hallucination policy: ambiguous data is tagged [**unclear -- verify**] rather than inferred; outputs mapped to HL7 FHIR standards ensuring full EHR interoperability.
- Architected a secure Python middleware proxy to prevent CORS vulnerabilities and ensure API keys are never exposed client-side; deployed in a sandboxed portable environment with no external dependencies.

Calder AI – Workflow Automation System

FastAPI, Sentence Transformers, Groq API, Docker, Python

github

- Built a production-ready AI workflow automation platform for intelligent customer support classification and automated response generation using zero-shot classification and LLM inference.

- Implemented NLP pipeline with sentiment analysis, urgency detection, and automated ticket processing; integrated fallback classifiers and structured logging for production reliability.
- Developed scalable FastAPI backend with analytics dashboard, CSV bulk processing endpoint, and automated testing suite; containerized with Docker for portable deployment.
- Designed responsive enterprise UI with real-time analytics; followed API validation best practices with comprehensive request/response logging.

BugSquash CI – AI-Powered QA Test Management Platform

Angular 17, Node.js, MongoDB, Gemini AI,

Socket.io, Docker

[github](#)

- Built an AI-first QA platform integrating Google Gemini 1.5 Flash to auto-generate full test cases (steps, expected results, priority) from a one-line title – significantly reducing test documentation time.
- Implemented Intelligent Failure Analysis: backend AI service analyzes CI/CD execution logs and suggests root cause and fix instantly, removing manual log digging.
- Delivered real-time metrics via Socket.io, PDF report export for stakeholders, automated email notifications, and enterprise security with JWT, RBAC, and account lockout; containerized with Docker and Nginx.

ClassEase – FAST University Timetable Management System

React.js, Node.js, Express.js, MongoDB, GitHub

Pages, Render

[github](#)

- Applied full software engineering lifecycle – requirements elicitation, SRS documentation, system design (UML use case, class, and sequence diagrams), implementation, and integration testing – for a system serving **2000+ students**.
- Architected RESTful backend with Node.js and Express.js; designed normalized MongoDB schema supporting real-time schedule filtering, conflict detection, and a role-based admin panel for timetable management.
- Led QA process achieving **100% pass rate** across 8 test suites using Katalon Recorder; managed Agile Scrum delivery over 3 sprints completing 25 user stories tracked in Trello.
- Deployed frontend on GitHub Pages and backend on Render; coordinated a 4-person team following version-controlled branching strategy on GitHub.

Achievements & Certifications

- **Bug Busters Coding Competition:** Winner among 20+ university teams (FAST University + Askari Bank) – demonstrated strong algorithmic problem-solving under competition pressure.
- **Academic Excellence:** Maintaining 3.5/4.0 CGPA at FAST NUCES while delivering production-grade AI projects and a paid teaching role simultaneously.
- **LeetCode:** Solved **200+ algorithmic problems** with consistent practice and strong DSA foundation.
- **Certifications:** Google AI Essentials, Postman API Fundamentals, Git/GitHub Fundamentals, Jira Fundamentals.

Additional

Languages: English (Professional), Urdu (Native)

Availability: Summer 2026 - AI Engineering, Generative AI, Full-Stack AI Development

Interests: AI Agents, LLM Systems, RAG Pipelines, Intelligent Automation, Competitive Programming

Activities: Coding Contest Participant, FAST Software Engineering Society